

DBMS CAPSTONE PROJECT REPORT

[ENCS204]

CAR RENTAL MANAGEMENT SYSTEM

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ABSTRACT

The Car Rental Management System is a comprehensive web-based application developed to streamline and automate the operations of a car rental service. The system is built using Flask as the backend framework and MySQL as the database, ensuring efficient data handling and smooth interaction between different system components.

In traditional rental systems, operations such as maintaining customer records, tracking vehicle availability, processing bookings, and managing payments are performed manually. These processes are time-consuming, inefficient, and prone to human errors. Moreover, manual systems lack real-time updates, which makes it difficult for administrators to manage multiple rentals simultaneously.

This project aims to eliminate these drawbacks by providing a fully automated and integrated system that ensures accuracy, reliability, and ease of use. The system enables users to add customers, manage vehicle records, rent cars, return vehicles with validation, and generate detailed invoices. It also incorporates a payment module that records the payment method used during the transaction, making the system more realistic and practical.

One of the key features of the system is the automatic calculation of rental cost based on the number of days, along with additional late fees if the vehicle is returned after the due date. The system also provides a structured invoice that includes customer details, car information, rental duration, payment method, and total cost, thereby enhancing transparency and professionalism.

Furthermore, the system maintains a centralized database, ensuring data integrity and consistency. It also supports real-time updates of car availability, which prevents double booking and improves overall system efficiency.

The Car Rental Management System serves as an effective solution for small to medium-scale rental businesses and demonstrates how modern web technologies and database management systems can be used to build scalable and efficient applications.

KEYWORDS: Car Rental System, Flask, MySQL, Automation, Billing System, Payment Integration, Database Management

1. INTRODUCTION

In today's digital era, the demand for efficient and automated systems has significantly increased across all industries, including transportation services. Car rental services have become a popular choice for individuals and businesses due to their flexibility, convenience, and cost-effectiveness. However, managing such services manually poses several challenges, especially when dealing with a large number of customers, vehicles, and transactions.

Traditional car rental systems rely heavily on paper-based records and manual operations. This approach leads to several issues such as data redundancy, lack of proper organization, difficulty in retrieving information, and increased chances of human errors. Additionally, tracking vehicle availability and managing bookings becomes complex, which affects overall service quality.

To address these challenges, the Car Rental Management System is developed as a web-based application that automates all major operations of a rental service. The system integrates customer management, car management, booking process, return handling, payment tracking, and invoice generation into a single platform.

The system is designed with a user-friendly interface that allows users to easily interact with the application. Administrators can efficiently manage car inventory, monitor rental activities, and maintain accurate records, while customers benefit from quick and reliable services.

By replacing manual processes with a computerized system, the application improves efficiency, reduces errors, and enhances data security. It also provides real-time updates, ensuring better decision-making and improved customer satisfaction.

2. MOTIVATION

The motivation behind developing the Car Rental Management System is to overcome the limitations of traditional rental systems and provide a modern, efficient, and reliable solution.

Manual systems are not suitable for handling large-scale operations due to their inefficiency and lack of automation. As the number of customers and vehicles increases, managing records manually becomes increasingly difficult.

Key Motivating Factors:

- Inefficiency of manual systems

Manual record-keeping consumes time and effort, making operations slow and unreliable.

- Data inconsistency and redundancy

Duplicate records and incorrect entries create confusion and reduce data accuracy.

- Lack of real-time updates

Manual systems cannot provide instant updates on car availability and booking status.

- Poor customer experience

Delays in processing bookings and payments affect customer satisfaction.

- Need for automation

Modern businesses require automated systems to improve productivity and performance.

The proposed system is designed to address these issues by providing automation, accuracy, and real-time data access.

3. LITERATURE REVIEW

A review of existing car rental systems shows that earlier solutions were primarily manual and lacked efficiency. With technological advancements, some systems introduced partial automation, but they still faced several limitations.

Many existing systems provide basic functionalities such as customer registration and car booking, but they lack advanced features like automated billing, payment tracking, and proper validation mechanisms. Additionally, some systems do not use well-structured databases, leading to data inconsistency and difficulty in managing large datasets.

Observations:

- Manual systems are outdated and inefficient
- Lack of integration between system components
- Poor user interface design
- Limited functionality in terms of billing and payment
- No real-time tracking of car availability

This project improves upon existing systems by integrating all functionalities into a single platform and providing advanced features such as payment handling, validation, and invoice generation.

4. GAP ANALYSIS

The analysis of existing systems reveals several gaps that need to be addressed:

- Lack of complete automation
- Absence of centralized database
- Inefficient booking and return process
- No proper billing and payment tracking
- Lack of validation for incorrect inputs
- Poor data security and reliability

The proposed system bridges these gaps by providing:

- Fully automated workflow
- Centralized and structured database
- Real-time updates
- Payment and billing system
- Input validation and error handling

5. PROBLEM STATEMENT

Traditional car rental systems are inefficient due to their dependence on manual operations and lack of automation. Managing customer records, vehicle availability, rental bookings, and payment transactions manually leads to various challenges such as data inconsistency, redundancy, and human errors.

In a growing rental business, it becomes difficult to track which cars are available, which are rented, and which have been returned. Additionally, calculating rental costs manually often leads to incorrect billing, especially when late return charges are involved.

Another major issue is the absence of a proper payment tracking system. Without a structured approach, it becomes difficult to maintain records of payment modes and total revenue generated. Furthermore, manual systems do not provide real-time updates, making decision-making slow and inefficient.

Therefore, there is a need for a computerized system that can:

- Automate all rental operations
- Maintain accurate and centralized records
- Provide real-time updates of car availability
- Reduce manual errors
- Ensure efficient billing and payment tracking

The proposed Car Rental Management System aims to solve these problems by providing an integrated, automated, and user-friendly solution.

6. OBJECTIVES

The primary objective of this project is to develop a robust and efficient system that simplifies the management of car rental operations.

The specific objectives are:

- To design and develop a web-based application using Flask and MySQL
- To maintain a centralized database for customers, cars, and rentals
- To automate the process of renting and returning cars
- To calculate rental cost dynamically based on duration
- To implement late fee calculation for delayed returns
- To integrate a payment system for recording transaction modes
- To generate detailed and professional invoices
- To ensure data validation and prevent invalid inputs
- To improve system efficiency, reliability, and user experience

The system aims to provide a seamless experience for both administrators and users while maintaining data accuracy and consistency.

7. Tools/Technologies Used

The system is developed using modern technologies that provide flexibility, efficiency, and scalability.

Programming Language: Python

Python is used for backend development due to its simplicity, readability, and extensive library support.

Framework: Flask

Flask is a lightweight web framework that allows easy development of web applications and integration with databases.

Frontend: HTML, CSS

HTML is used for structuring web pages, while CSS is used for styling and creating an attractive user interface.

Database: MySQL

MySQL is used as the database to store and manage all system data efficiently. It provides strong support for relational data and ensures data integrity.

IDE: Visual Studio Code

VS Code is used for coding, debugging, and managing the project efficiently.

8.SYSTEM ARCHITECTURE

The system follows a three-tier architecture, which separates the application into different layers for better organization and scalability.

1. Presentation Layer

This layer handles the user interface using HTML and CSS. It allows users to interact with the system through forms and displays.

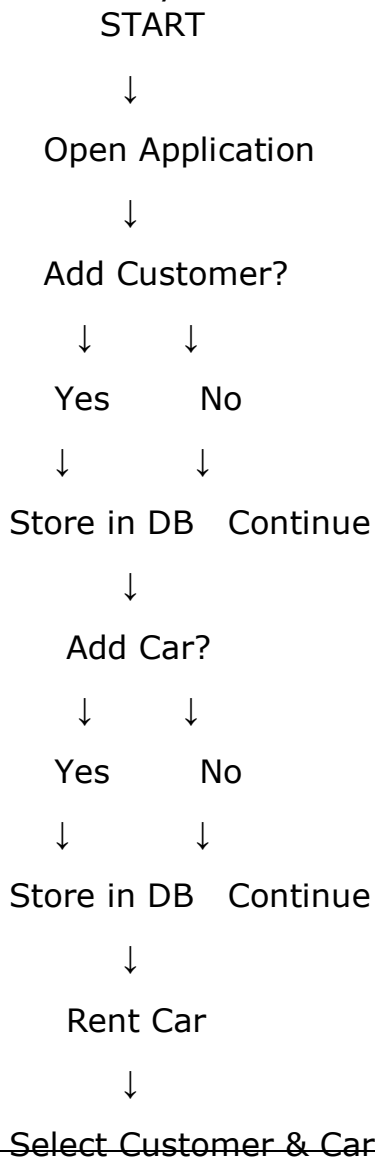
2. Application Layer

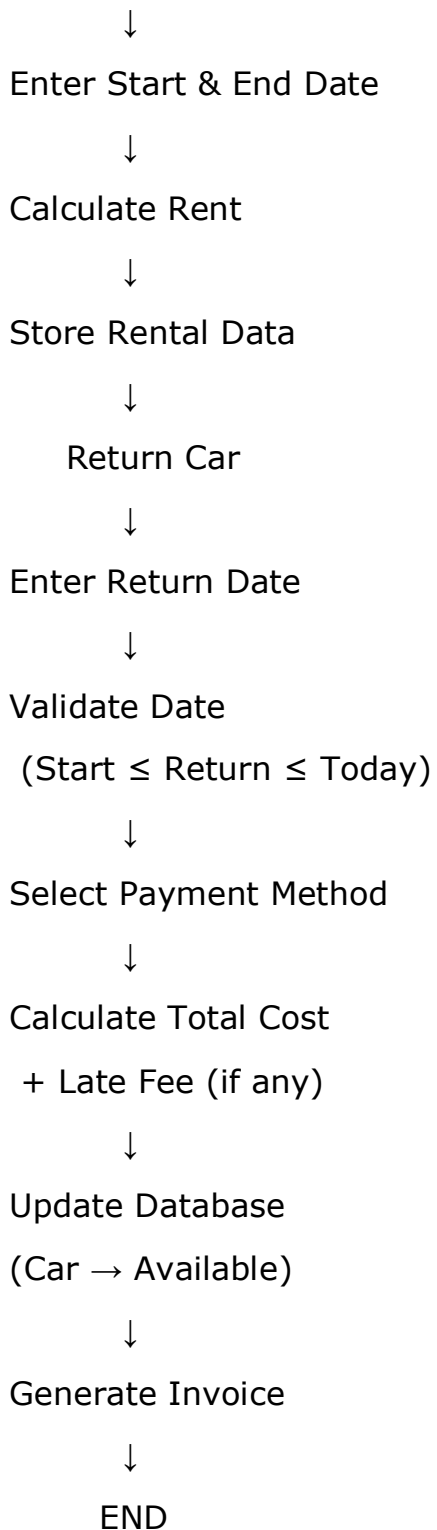
This layer is developed using Flask. It processes user requests, performs calculations, handles business logic, and communicates with the database.

3. Database Layer

This layer uses MySQL to store and retrieve data. It maintains records of customers, cars, rentals, and payments.

Flow of System:





This architecture ensures better performance, maintainability, and scalability of the system.

9. METHODOLOGY

The development of the system follows a structured approach to ensure proper implementation of all functionalities.

Step 1: Customer Management

Users can add customer details such as name, email, and phone number. These details are stored in the database.

Step 2: Car Management

Cars are added with details such as brand, model, and price per day. Each car has a status (Available/Rented).

Step 3: Rental Process

Users select a customer and an available car. The system records the start and end date and calculates the estimated cost.

Step 4: Return Process

Users enter the return date and select the payment method. The system validates the return date to ensure it is within the allowed range.

Step 5: Cost Calculation

The system calculates total cost based on number of days and adds late fee if the car is returned after the due date.

Step 6: Payment Handling

The selected payment method (Cash, Online, Card) is stored in the database.

Step 7: Invoice Generation

A detailed invoice is generated showing all relevant information including customer details, car details, rental duration, payment method, and total cost.

Step 8: Database Update

The system updates the car status to "Available" after return and stores all transaction data.

10.CONCLUSION

The Car Rental Management System is a comprehensive and efficient solution designed to automate the operations of a car rental business. The system successfully integrates customer management, car management, rental processing, payment handling, and invoice generation into a single platform.

By replacing manual processes with an automated system, the application improves efficiency, accuracy, and reliability. The use of modern technologies such as Flask and MySQL ensures scalability and flexibility.

Overall, the project demonstrates how database management systems and web technologies can be effectively combined to solve real-world problems. It provides a strong foundation for further enhancements and serves as a valuable learning experience in software development and database management.

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